

```
# Install necessary libraries: transformers for NLP models, datasets for data loading, torch for
!pip install transformers datasets torch scikit-learn
```

Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (1.6.1)

Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers)

Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers)

Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers)

Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers)

Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers)

Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers)

Requirement already satisfied: tokenizers<0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers)

Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (from transformers)

Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers)

Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers)

Requirement already satisfied: pyarrow>=15.0.0 in /usr/local/lib/python3.12/dist-packages (from datasets)

Requirement already satisfied: dill<0.3.9,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from datasets)

Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (from datasets)

Requirement already satisfied: requests>=2.32.2 in /usr/local/lib/python3.12/dist-packages (from datasets)

Requirement already satisfied: xxhash in /usr/local/lib/python3.12/dist-packages (from datasets)

Requirement already satisfied: multiprocessing<0.70.17 in /usr/local/lib/python3.12/dist-packages (from datasets)

Requirement already satisfied: fsspec<=2025.3.0,>=2023.1.0 in /usr/local/lib/python3.12/dist-packages (from datasets)

Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from datasets)

Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch)

Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch)

Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch)

Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch)

Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from torch)

Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from torch)

Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from torch)

Requirement already satisfied: aiohttp!=4.0.0a0,!4.0.0a1 in /usr/local/lib/python3.12/dist-packages (from torch)

Requirement already satisfied: hf-xet<2.0.0,>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from transformers)

Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from transformers)

Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from huggingface-hub)

Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests)

Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests)

Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests)

Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests)

Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy)

Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2)

Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas)

Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas)

Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas)

Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp)

Requirement already satisfied: aiosignal>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp)

Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp)

Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from aiohttp)

Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.12/dist-packages (from aiohttp)

Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp)

Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp)

Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx)

Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx)

Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore)

Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil)

Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer)

Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer)

Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer)

Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer)

Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich)

Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich)

Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py)

```
# Import the PyTorch library, a deep learning framework.
import torch
# Import the NumPy library for numerical operations.
import numpy as np
# Import load_dataset from the datasets library to easily load public datasets.
from datasets import load_dataset
# Import AutoTokenizer and AutoModelForSequenceClassification from transformers for model and to
from transformers import AutoTokenizer, AutoModelForSequenceClassification
# Import TrainingArguments and Trainer for defining training configurations and managing the tra
from transformers import TrainingArguments, Trainer
# Import accuracy_score and precision_recall_fscore_support from scikit-learn for evaluating mod
from sklearn.metrics import accuracy_score, precision_recall_fscore_support
```

```
# Load the 'imdb' dataset, a popular dataset for sentiment analysis.
dataset = load_dataset("imdb")

# Split the dataset into training and testing sets.
train_dataset = dataset['train']
test_dataset = dataset['test']

# Print the first example from the training dataset to understand its structure.
print(train_dataset[0])
```

```
{'text': 'I rented I AM CURIOUS-YELLOW from my video store because of all the controversy that su
```

```
# This cell is an output displaying the structure of a single example from the IMDb dataset.
# It shows a 'text' field containing the movie review and a 'label' field indicating sentiment (
{
  'text': 'This movie was fantastic...',
  'label': 1
}
```

```
{'text': 'This movie was fantastic...', 'label': 1}
```

```
# Load a pre-trained tokenizer for the 'distilbert-base-uncased' model.
# The tokenizer converts text into numerical input IDs and attention masks suitable for the mode
tokenizer = AutoTokenizer.from_pretrained("distilbert-base-uncased")
```

```
# Define a function to tokenize the text data.
# It applies the loaded tokenizer, padding sequences to a maximum length and truncating longer o
def tokenize_function(example):
    return tokenizer(example["text"], padding="max_length", truncation=True)

# Apply the tokenization function to both the training and testing datasets.
# `batched=True` processes multiple examples at once for efficiency.
train_dataset = train_dataset.map(tokenize_function, batched=True)
test_dataset = test_dataset.map(tokenize_function, batched=True)
```

```
Map: 100% 25000/25000 [01:28<00:00, 334.59 examples/s]
```

```
# Set the format of the datasets to 'torch' tensors.
# This also specifies which columns ('input_ids', 'attention_mask', 'label') should be used by t
train_dataset.set_format(type='torch', columns=['input_ids', 'attention_mask', 'label'])
test_dataset.set_format(type='torch', columns=['input_ids', 'attention_mask', 'label'])
```

```
# Load a pre-trained DistilBERT model for sequence classification.
# `num_labels=2` specifies that the model should output probabilities for 2 classes (positive/ne
model = AutoModelForSequenceClassification.from_pretrained(
    "distilbert-base-uncased",
    num_labels=2
)
```

Loading weights: 100% 100/100 [00:00<00:00, 218.04it/s, Materializing param=distilbert.transformer.layer.5.sa_layer_norm.]

DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased

Key	Status
vocab_layer_norm.bias	UNEXPECTED
vocab_projector.bias	UNEXPECTED
vocab_layer_norm.weight	UNEXPECTED
vocab_transform.bias	UNEXPECTED
vocab_transform.weight	UNEXPECTED
pre_classifier.weight	MISSING
classifier.weight	MISSING
pre_classifier.bias	MISSING
classifier.bias	MISSING

Notes:

- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect
- MISSING :those params were newly initialized because missing from the checkpoint. Consider tr

```
# Define a function to compute evaluation metrics (accuracy, F1-score, precision, recall).
# This function will be called during evaluation to assess the model's performance.
def compute_metrics(pred):
    # Extract true labels from the prediction object.
    labels = pred.label_ids
    # Get predicted labels by taking the argmax of the model's output logits.
    preds = pred.predictions.argmax(-1)

    # Calculate precision, recall, and F1-score. 'average='binary' is suitable for binary classif
    precision, recall, f1, _ = precision_recall_fscore_support(labels, preds, average='binary')
    # Calculate the accuracy score.
    acc = accuracy_score(labels, preds)

    # Return a dictionary of computed metrics.
    return {
        'accuracy': acc,
        'f1': f1,
        'precision': precision,
        'recall': recall
    }
```

```
# Configure training arguments for the Trainer.
# This includes specifying the output directory, learning rate, batch sizes, number of epochs, and
training_args = TrainingArguments(
    output_dir="./results",
    learning_rate=2e-5,
    per_device_train_batch_size=8,
    per_device_eval_batch_size=8,
    num_train_epochs=2,
    weight_decay=0.01
)
```

```
# Initialize the Hugging Face Trainer with the model, training arguments, datasets, and custom met
trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=train_dataset,
    eval_dataset=test_dataset,
    compute_metrics=compute_metrics
)
```

```
# Start the training process.
```

```
trainer.train()
```

```
/usr/local/lib/python3.12/dist-packages/torch/utils/data/dataloader.py:775: UserWarning: 'pin_mem
super().__init__(loader)
```

[9/6250 03:41 < 54:49:42, 0.03 it/s, Epoch 0.00/2]

Step Training Loss

```
# Evaluate the trained model on the evaluation dataset.
# This will run the `compute_metrics` function and return a dictionary of evaluation results.
results = trainer.evaluate()
# Print the evaluation results.
print(results)
```

